TOPIC 4: INTRODUCTORY ORGANIC CHEMISTRY AND ALKANES

GRADE 11

SEMESTER 1/2020

Checklist

4A INTRODUCTION TO ORGANIC CHEMISTRY ✓

Students will be assessed on their ability to:

4.1 understand the difference between hazard and risk ☐

4.2 understand the hazards associated with organic compounds and why it is necessary to carry out risk assessments when dealing with potentially hazardous materials ☐

4.3 be able to suggest ways in which risks can be reduced and reactions carried out safely, for example:

i working on a smaller scale ☐

ii taking precautions specific to the hazard ☐

iii using an alternative method that involves less hazardous substances ☐

4.4 understand the concepts of homologous series and functional group ☐

4.5 be able to apply the rules of International Union of Pure and Applied Chemistry (IUPAC) nomenclature to:

i name compounds relevant to this specification ☐

ii draw these compounds, as they are encountered in the specification, using structural, displayed and skeletal formulae Students will be expected to know prefixes for compounds up to C10 ☐

4.6 be able to classify reactions as addition, substitution, oxidation, reduction or polymerization ☐

4.7 understand that bond breaking can be:

i homolytic, to produce free radicals ☐

ii heterolytic, to produce ions ☐

4.8 know definitions of the terms ‘free radical’ and ‘electrophile’ ☐

4.9 know the general formula of alkanes and cycloalkanes, and understand that they are hydrocarbons (compounds of carbon and hydrogen only) which are saturated (contain single bonds only) ☐

4.10 understand the term ‘structural isomerism’ and be able to draw the structural isomers of organic molecules, given their molecular formula ☐

4.11 be able to draw and name the structural isomers of alkanes and cycloalkanes with up to six carbon atoms ☐

4.12 know that alkanes are used as fuels and obtained from the fractional distillation, cracking and reforming of crude oil, and be able to write equations for these reactions ☐

4.13 know that pollutants, including carbon monoxide, oxides of nitrogen and sulfur, carbon particulates and unburned hydrocarbons, are emitted during the combustion of alkane fuels ☐

4.14 understand the problems arising from pollutants from the combustion of alkane fuels, limited to the toxicity of carbon monoxide and why it is toxic, and the acidity of oxides of nitrogen and sulfur ☐

4.15 be able to discuss the reasons for developing alternative fuels in terms of sustainability and reducing emissions, including the emission of CO2 and its relationship to climate change ☐

4.16 be able to apply the concept of carbon neutrality to different fuels, such as petrol, bioethanol and hydrogen ☐

4.17 understand the reactions of alkanes with:

i oxygen in the air (combustion) ☐

ii halogens ☐

4.18 understand the mechanism of the free radical substitution reaction between an alkane and a halogen:

i using free radicals, which are species with an unpaired electron, represented by a single dot ☐

ii showing the initiation step of the mechanism, with curly half-arrows for free radical formation ☐

iii showing the propagation and termination steps of the mechanism ☐

iv having limited use in synthesis because of further substitution reactions ☐

Further suggested practical Cracking alkanes by thermal decomposition, including liquid paraffin using aluminium oxide as a catalyst

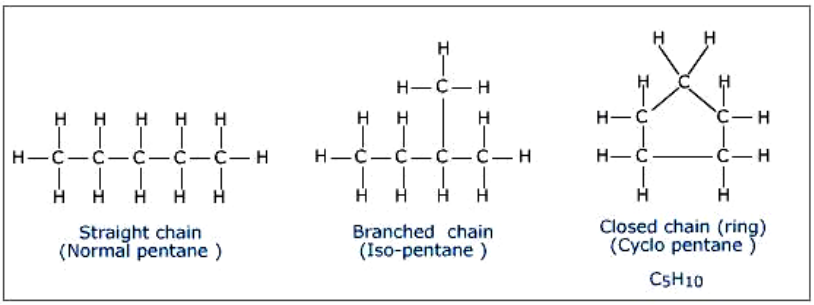
ORGANIC CHEMISTRY

What is organic chemistry?

- Organic chemistry is the chemistry of carbon compounds (except few compounds of carbon such as oxides, carbonates and chlorides).
- These organic compounds are the main constituents of all animal and plant life.

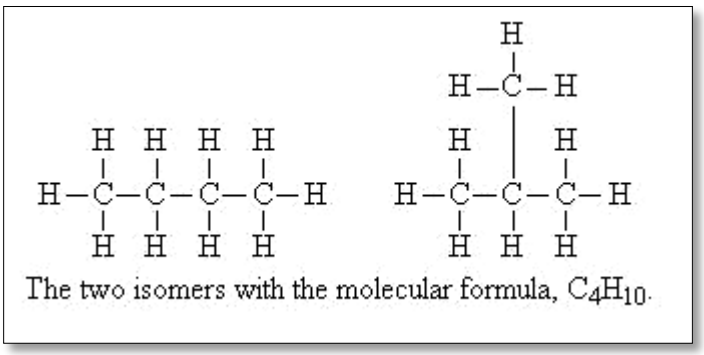
1. Catenation

- Catenation is the ability to form bonds between the atoms of the same element.
- Carbon exhibits catenation to maximum extent because of strong carbon-carbon bond and tetracovalency.
- Due to this property of catenation, carbon atoms can form various types of straight chain, branched chain and ring structures, thus giving rise to a large variety of compounds.

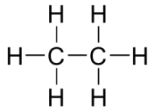


2. Formation of isomers

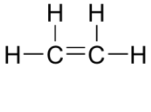
- Organic compounds with the same molecular formula can have two or more different structures.
- This phenomenon is known as isomerism.
- This property of carbon is also responsible for formation of such a large number of organic compounds.


The two isomers with the molecular formula, C_4H_{10}.

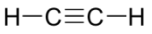
3. ***Formation of multiple bonds***
    - Carbon atom can easily form multiple bonds (double or triple) with carbon, oxygen, nitrogen, etc.



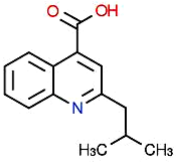
ethane
(an alkane)



ethene
(an alkene)



ethyne
(an alkyne)



4.1 HAZARD AND RISK

- ***Hazard:*** Potential of a substance to do harm.
- Example: some chemicals are flammable and some are toxic. A hazard associated with a chemical will always be the same.
- ***Risk***: the chance that a particular hazardous substance will actually cause harm.
- Example: petrol is flammable, but the risk of catching fire depends on the environment in which it is used. The risk of coming to harm when filling a car is very small. But the risk of harm is very high if it is exposed to fire. The risk of harm depends on the environment in which the substance is used and how it is handled.

4.2 Hazards associated with Organic chemicals

- Many organic chemicals are
    - Poisonous
    - Carcinogenic
    - Flammable
- As a result, it is very important to carry out a full risk assessment to identify the risk of using these hazardous chemicals and to reduce the risks as far as possible.

4.3 Ways of reducing risk

- *Working on a smaller scale.*Eg: there is less risk of inhaling fumes with smaller amount of chemicals.
- *taking specific precautions or using alternative techniques depending on the properties of the hazardous substance you are using:* Eg: Use the lowest possible concentration so that the risk is reduced.
- *Careful use of safety measures:* Eg: Use fume cupboards to remove toxic fumes, wear gloves and safety goggles.
- *carrying out the reaction using an alternative method that involves less hazardous substances.*Eg: Use a substitute less hazardous chemical to study the same basic reaction.
- *Changing the conditions under which the reaction takes place:* Eg: Lower the temperature of a reaction mixture so that the rate of reaction is reduced. This in turn reduces the risks associated with the chemicals

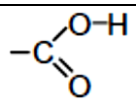
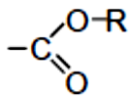
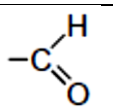
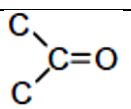
4.4 Homologous series

- The vast number of organic compounds can be arranged into a relatively small number of groups or families known as homologous series.
- A homologous series is a series of compounds with the ***same functional group***, the ***same general formula***, and each member ***differs from the next by a CH_2 group.***
- All share the same general formula
- Formula differs from their neighbours by CH_2.
  Eg: CH_4, C_2H_6 ,C_3H_8…..etc
- Contain the same functional group(s).
- Have similar chemical properties.
- Show a gradual change in physical properties as molar mass increases.
- Can usually be prepared by same methods.

4.4 Functional Groups

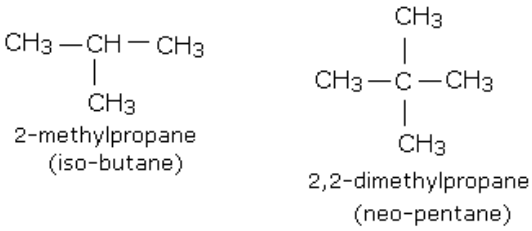
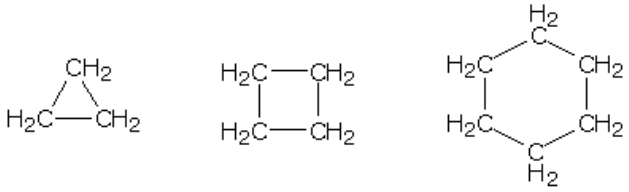
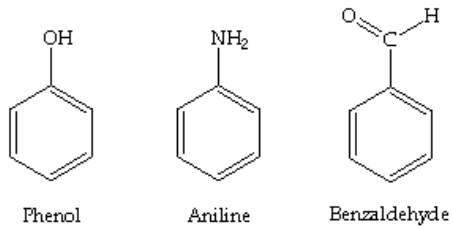
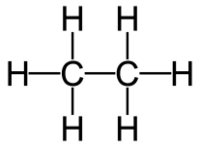
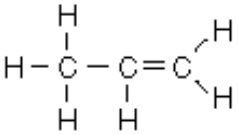
- A group of atoms responsible for the characteristic reactions of a compound.
- A functional group can consist of
  - one atom **— Br**
  - a group of atoms **— OH**
  - multiple bonds between carbon atoms **C=C**
- Each functional group has its own distinctive properties.

SOME COMMON FUNCTIONAL GROUPS

GROUP	ENDING	GEN. FORMULA	STRUCTURE	EXAMPLE
ALKANE	- ane	RH	C – C	C_2H_6 ethane
ALKENE	- ene		C = C	C_2H_4 ethene
HALOALKANE	Halo	- RX	C – Cl	C_2H_5Cl chloroethane
ALCOHOL	- ol	ROH	–O–H	C_2H_5OH ethanol
CARBOXYLIC ACID	-oic acid	RCOOH		CH_3COOH ethanoic acid
ESTER	- yl - oate	RCOOR		CH_3COOCH_3 methylethanoate
ALDEHYDE	-al	RCHO		CH_3CHO ethanal
KETONE	- one	RCOR		CH_3COCH_3propanone

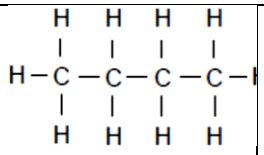
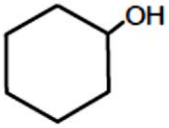
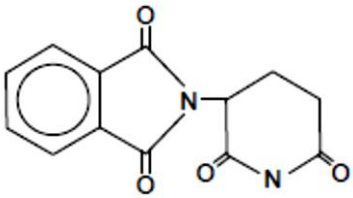
The symbol R represents groups of carbon and hydrogen atoms in the rest of the molecule

4.5 Organic Families – Classifying organic compounds

Aliphatic	Molecules contain straight- or branched- chain skeletons.	
Alicyclic	Molecules consist of closed rings of carbon atoms which may contain single or multiple carbon-carbon bonds.	
Aromatic or Arenes	Compounds containing atleast one benzene ring.	
Saturated	Molecules containing only single bonds.	
Unsaturated	Molecules containing double bonds or triple bonds.	

Use of different formulae in organic chemistry

General	the simplest algebraic formula for a member of a homologous series	C_nH_{2n+2} C_nH_{2n}	for an alkane for an alkene
Molecular	shows the exact number of atoms of each element in a molecule	C_4H_{10}	for butane
Empirical	shows the simplest whole number ratio of atoms of each element in a molecule	C_2H_5	for butane
Structural	the minimal detail that shows the arrangement of atoms in a molecule	$CH_3CH_2CH_2CH_3$ $CH_3CHOHCH_3$	Butane propan-2-ol

Displayed Structural	shows the relative positioning of atoms and the number of bonds between them		butane
Skeletal	used to show a simplified organic formula by removing hydrogen atoms from alkyl chains, leaving just a carbon skeleton and cyclohexane associated functional groups. • each covalent bond is shown by a line • a carbon atom is at the join of lines • functional groups are shown • the number of hydrogen atoms on each carbon atom is the difference between the number of lines and 4	   	Cyclohexane cyclohexene cyclohexanol butan-1-ol
Space filling models	Skeletal formulae tend to be used with larger organic molecules - e.g. *thalidomide*  Folic acid molecule  Used to model a molecule in three Dimensions.		

4.5 Nomenclature in organic chemistry

IUPAC stands for – International Union of Pure and Applied Chemistry

Systems A naming system must tell you everything about a structure without ambiguity.

There are two types of naming system commonly found in organic chemistry;

Trivial :
- based on some property or historical aspect;
- the name tells you little about the structure

Systematic:
- based on an agreed set of rules (I.U.P.A.C);
- exact structure can be found from the name (and vice versa)

Series	***trivial name***	***systematic name***	***example(s)***
	Paraffin	alkane	methane, butane
	Olefin	alkene	ethene, butane
	fatty acid	alkanoic (carboxylic) acid	ethanoic acid
Compounds	***trivial name***	***systematic name***	
	acetic acid	ethanoic acid (CH_3COOH)	

Systematic (IUPAC) Nomenclature

STEM Shows the number of carbon atoms in longest chain bearing the functional group and (if necessary) a prefix showing the position and identity of any substituents.

PREFIX Apart from the first four, which retain trivial names, the number of carbon atoms is indicated by a prefix derived from the Greek numbering system.

- the list of alkanes demonstrate the use of prefixes
- the ending is the same as they are all alkanes

Prefix	*C atoms*	*Alkane*
meth-	1	methane
eth-	2	ethane
prop-	3	propane
but-	4	butane
pent-	5	pentane
hex-	6	hexane
hept-	7	heptane
oct-	8	octane
non-	9	nonane
dec-	10	decane

The **ending** tells you **which functional group** is present

SUFFIX

If any functional groups are present, add relevant ending to the basic stem. The position of the functional group must be given to avoid any confusion.

Suffix	Homologous series
-ane	Alk**ane**
-ene	Alk**ene**
-ol	Alcoh**ol**
-oic	Carboxyl**ic** acid
-al	**Al**dehyde
-one	Ket**one**

SUBSTITUENTS Many compounds have substituents (additional atoms, or groups of atoms) attached to the chain. Their position is numbered according to a set of rules.

LOCANT Number which indicates where a side chain or functional group is attached. Example: **2**-chlorobutane, but-**2**-ene

NOMENCLATURE - GENERAL RULES

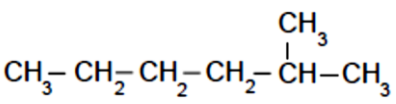
Stem

- look for the longest chain of carbon atoms containing the functional group.
- the carbon atoms must be in a continuous row.
- use prefixes as shown on page 9
- ending tells you what type of carbon structure you have; alkanes end in **ANE**

Side-chain

- carbon based substituents are named before the chain name.
- they have the prefix -yl added to the basic stem (e.g. CH_3 is methyl).
- Number the principal chain from one end so that the side chain is attached to a carbon atom with the **lowest possible number**.

e.g. 2-methylhexane

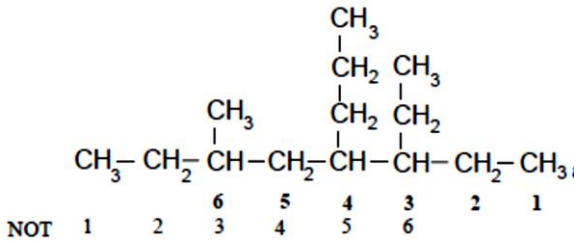


If there is more than one side-chain the following rules apply: -

- side-chain names appear in alphabetical order i.e. butyl, ethyl, methyl, propyl.
- number the principal chain from one end to give the lowest numbers.
- each side-chain is given its own number.
- if identical side-chains appear more than once, prefix with di, tri, tetra etc
- numbers are separated from names by a HYPHEN ***2-methylheptane***
- numbers are separated from numbers by a COMMA ***2,3-dimethylbutane***

Example:

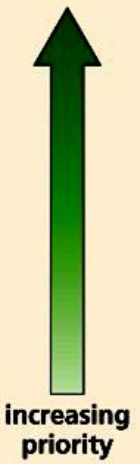
- *longest chain 8 (it is an octane)*
- *3,4,6 are the numbers NOT 3,5,6*
- *order is ethyl, methyl, propyl*



3-ethyl-6-methyl-4-propyloctane

If there is more than one functional group in a molecule: -

- a priority order list is considered and the group higher in the list is called the **principle functional group**. The suffix of the molecule is related to it and if any other functional group is present it is considered as a branch and given a prefix.
- The priority order of the functional groups can be listed as below. (Included only which required for GCE A/L)

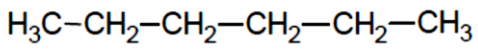
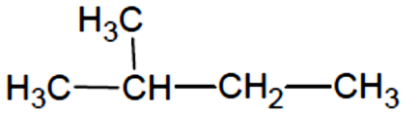
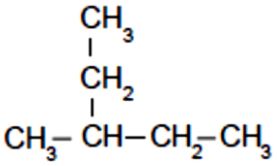
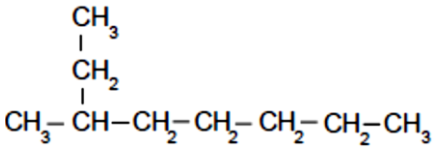
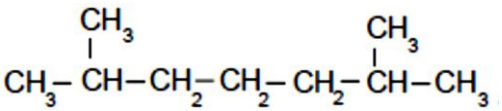
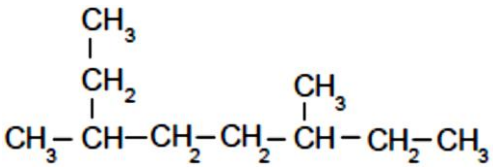


Class	Suffix Name	Prefix Name
Carboxylic acid	-oic acid	Carboxy
Ester	-oate	Alkoxycarbonyl
Amide	-amide	Amido
Nitrile	-nitrile	Cyano
Aldehyde	-al	Formyl (—CH=O)
Ketone	-one	Oxo (=O)
Alcohol	-ol	Hydroxy
Amine	-amine	Amino
Alkene	-ene	Alkenyl
Alkyne	-yne	Alkynyl
Alkane	-ane	Alkyl
Ether	—	Alkoxy
Alkyl halide	—	Halo

ALKANES

- Identify the longest chain of carbon atoms and use prefix according to the number of carbon atoms present in it.
- Use –***ane*** as the suffix.
- If there are branches, name side chain before the chain name and give a locant to the side chain with the lowest possible number.

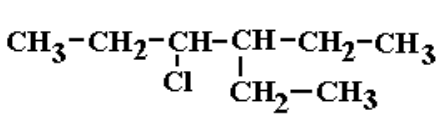
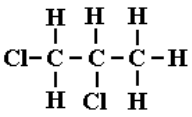
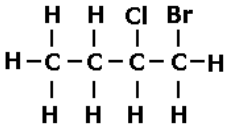
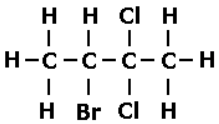
Exercise 1: Name the following alkanes

<table>
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  </td>
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<tr>
<td>2.
  </td>
<td>3.
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</tr>
<tr>
<td>4.
  </td>
<td>5.
  </td>
</tr>
</table>

HALOGENO ALKANES

- Identify the longest chain of carbon atoms containing the halogen.
- Halogen substituents are indicated by the prefixes fluoro, chloro, bromo and iodo, and are listed in alphabetical order with other substituents.
- Locant is used to show where the halogen atom is present.

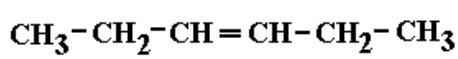
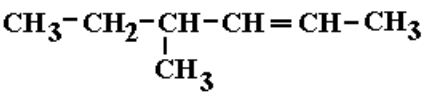
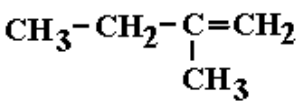
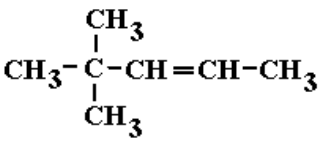
Exercise 2: Name the following halogenoalkanes

1. 	2. 
3. 	4. 

ALKENES

- Identify the longest chain of carbon atoms containing the functional group, C=C.
- The principle chain is not always the longest chain.
- The position of C=C is indicated by the lower numbered carbon atom on one end of the double bond. Count from one end to give the lowest number.
- Position of side chain is based on the number allocated to the C=C bond

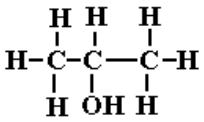
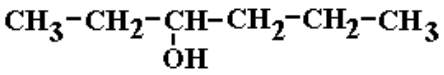
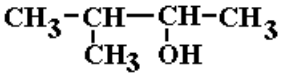
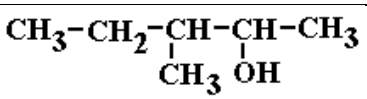
Exercise 3: Name the following alkenes

1. 	2. 
3. 	4. 

ALCOHOLS

- Identify the longest chain of carbon atoms containing the functional group, O—H.
- The principle chain is not always the longest chain.
- The position of O—H is indicated by the lowest possible number from one end.
- Position of side chain is based on the number allocated to the O—H.

Exercise 4: Name the following alcohols

1. 	2. 
3. 	4. 

Exercise 5: Draw the displayed structure and skeletal formula.

1. Butane	**2. 3-bromopentane**
3. 4-ethylpentane	**4. 2,3-dimethyloctane**
5. 3-methylhept-2-ene	**6. 2-chlorohex-3-ene**

7. Butan-2-ol	**8. Propanol**
9. 2-methylpropan-2-ol	**10. 3-chloro-2-methylpentan-3-ol**

4.6 TYPES OF ORGANIC REACTIONS

1. Addition reaction

- An addition reaction occurs when two substances react together to form a single substance.
- Most of the reactions of alkenes are addition reactions.
  Example: Reaction between ethene and bromine.

$$C_2H_4 + Br_2 \longrightarrow CH_2BrCH_2Br$$

2. Substitution reaction

- It is a reaction in which one atom or a group of atoms is replaced by another atom or group of atoms.
- Saturated organic molecules usually undergo substitution reactions.
  Example: Reaction between methane and chlorine in sunlight.

$$CH_4 + Cl_2 \longrightarrow CH_3Cl + HCl$$

3. Polymerization

- Polymerization occurs when thousands of monomer molecules join together to form a long polymeric chain.
- This long chain is known as the polymer.
  Example: polymerization of ethene

$$(C_2H_4)_n \longrightarrow -(CH_2\text{-}CH_2)_n-$$

4. Reduction

- When referred to organic molecules, it is the addition of hydrogen
  Example: Reduction of ketones to alcohols

$CH_3COCH_3 \longrightarrow CH_3CH(OH)CH_3$
Propanone propan-2-ol

5. Combustion reaction

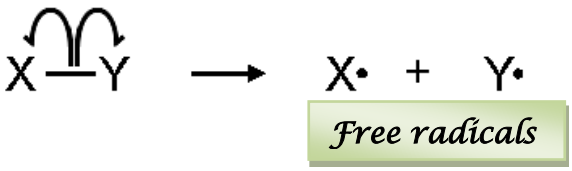
- It is a reaction with oxygen.
- Complete combustion of hydrocarbons produces carbon dioxide and water.
- Incomplete combustion of hydrocarbons produces carbon monoxide and water. Soot particles are also formed.
  Example:
  Complete combustion of propane: $C_3H_8 + 5O_2 \longrightarrow 3CO_2 + 4H_2O$
  Incomplete combustion of propane: $C_3H_8 + {}^7/_2O_2 \longrightarrow 3CO + 4H_2O$

4.7 BOND BREAKING IN ORGANIC REACTIONS

- Chemical reactions involve breaking and making of bonds between atoms.
- Covalent bonds in organic molecules can be broken by ***homolytic fission*** or ***heterolytic fission***.

1. Homolytic fission

- It is the method of splitting a covalent bond into two equal halves such that each species get one electron.
- This usually happens when the electronegativity of the two bonded atoms are same, which results in equal attraction by the two atoms to the electrons.
- Homolytic fission results in production of **free radicals**.



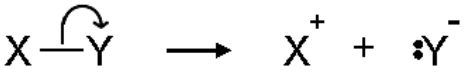
- A half headed curly arrow represents the movement of a single electron.

2. Heterolytic fission

- It is the unequal fission of the covalent bond.
- This happens when there is an electronegativity difference between the two bonded atoms.
- The atom having higher electronegativity attracts the pair of electrons more than the other atom.
- Therefore, when the bond breaks the higher electronegative atom gets the two electrons and the other atom does not get any electron.

- Heterolytic fission results in formation of positively charged species called electrophiles and negatively charged species called nucleophiles.

If the electronegativity of Y is greater than X, Y gets the pair of electrons.



X^+ is the electrophile / Y^- is the nucleophile

If the electronegativity of X is greater than Y, X gets the pair of electrons.



Y^+ is the electrophile / X^- is the nucleophile

- A full headed curly arrow represents the movement of a pair of electrons.

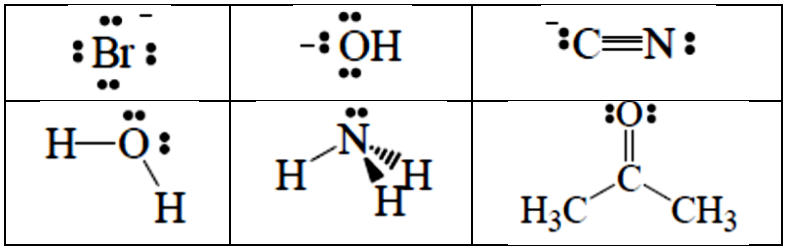
4.8 TYPE OF REAGENTS

1. Free radical

- It is an uncharged species with a single unpaired electron that is used to form a covalent bond.
- They are very reactive.
  Example: Cl•

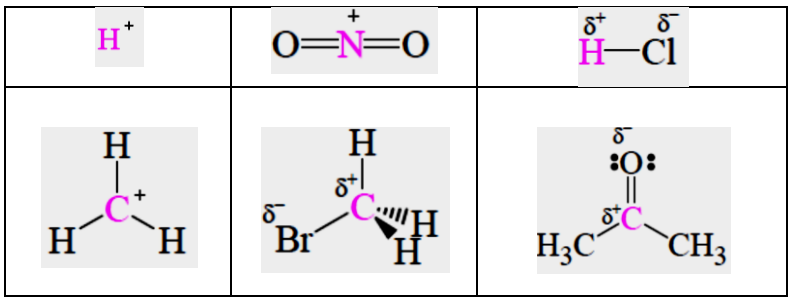
2. Nucelophile

- It is a species with a lone pair of electrons that is used to form a covalent bond with a δ+ atom in another molecule.
- It attacks electron deficient areas.
  Examples:



3. Electrophile

- It is a positively charged or a neutral species that bond to an electron rich site in a molecule.
- It accepts a pair of electrons from that site forms a new covalent bond.
  Examples:



- These neutral species can be polarized so that they can act as electrophiles.

Br—Br Cl—Cl I—I

REACTION MECHANISM

- Reaction type or **mechanism** describes the way in which the electron cloud of the covalent bond shifts during the reaction. These are classified according to:
  - **The type of bond fission**: homolytic OR heterolytic
  - **The nature of reagent molecule**, e.g. free radical, electrophile or nucleophile.
  - **The type of structural change: a**ddition, substitution, elimination, etc..

- Therefore a **reaction mechanism** can be classified as **heterolyticelectrophilc addition**, **heterolyticnucleophilic substitution** OR **homolytic free radical substitution**, etc.
- The types of reactions can be summarized as below

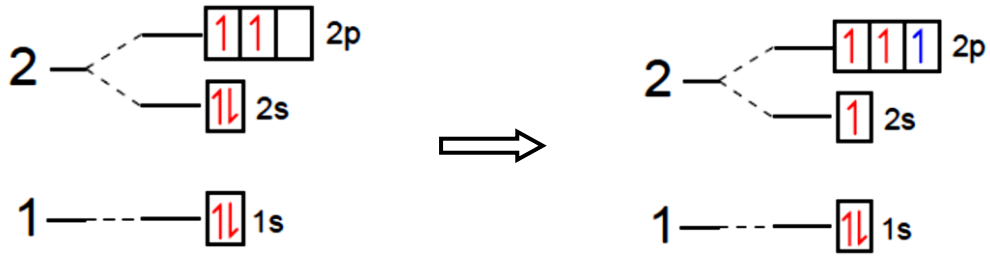
	Type of reaction		
Type of reagent	**Addition**	**Substitution**	**Elimination**
Nucleophile	✓	✓	✓
Electrophile	✓	✓	-
Free radical	✓	✓	-

4B ALKANES

4.9 Introduction:

- Alkanes are hydrocarbons that belongs to the homologous series with general formula C_nH_{2n+2} – (*non-cyclic only*)
- They are **saturated hydrocarbons.** It means all the bonds in alkanes are single bonds.
- They are highly flammable.
- Carbon atoms in alkanes are sp^3 **hybridized**.

Hybridization in carbon

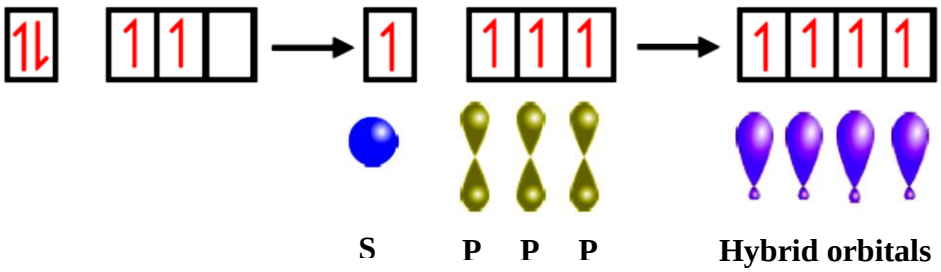


The electronic configuration of a carbon atom is $1s^2 2s^2 2p^2$

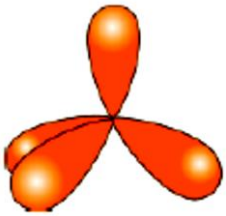
It is energetically stable to promote one electron from 2s orbital to the empty p orbital.

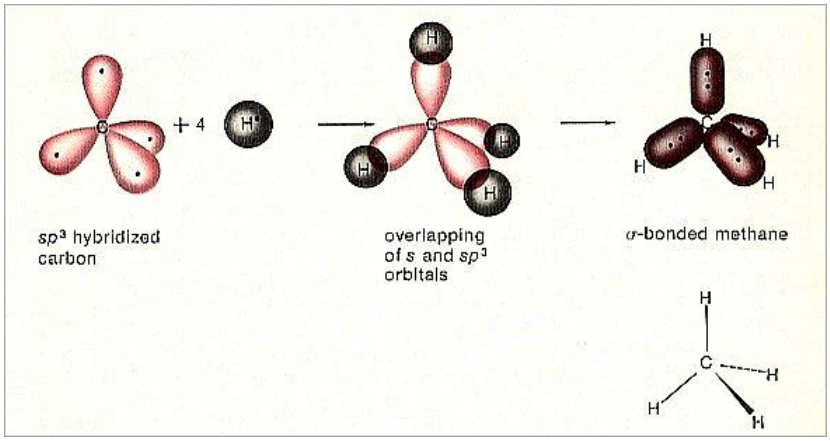
The configuration is now $1s^2 2s^1 2p^3$

- The process is **favourable** because of the arrangement of electrons; four unpaired means **less repulsion** and therefore **more stability**.



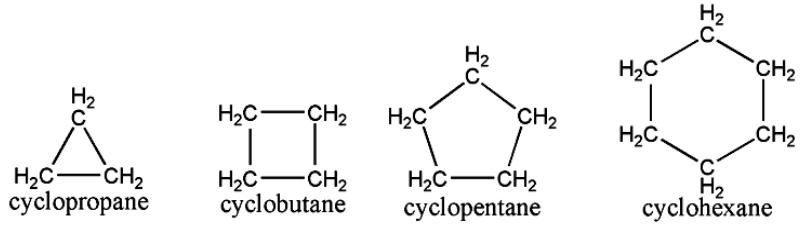
- The four orbitals (an s and three p's) combine or HYBRIDIZE to give four new orbitals. All four orbitals are equivalent.
- In ALKANES, the four sp^3 orbitals repel each other into a tetrahedral arrangement.
- Therefore, bonds are **spaced tetrahedrally** around carbon atoms.
- In alkanes, carbon forms **sigma bonds** with other carbons and hydrogens.





Cyclic or cyclo alkanes

- Cyclic alkanes are alkanes where the ends of the carbon chain have been linked to form a ring structure.
- Their general formula is not the same as simple alkanes.
- Their general formula is the same as alkenes, $\mathbf{C_nH_{2n}}$, because two hydrogens are lost when the ends of the chain are linked.



Nomenclature

In the **IUPAC** system,

- Identify the longest chain of carbon atoms.
- The carbon atoms must be in a continuous row.
- Use prefix according to the number of carbon atoms present in it
- Use –***ane*** as the suffix.

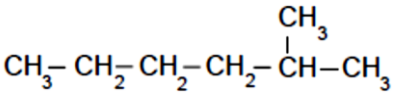
Name	Number of carbon atoms	Compound
Methane	1	CH_4
Ethane	2	CH_3CH_3
Propane	3	$CH_3CH_2CH_3$
Butane	4	$CH_3CH_2\ CH_2CH_3$
Pentane	5	$CH_3CH_2\ CH_2\ CH_2CH_3$
Hexane	6	$CH_3CH_2\ CH_2\ CH_2CH_2CH_3$
Heptane	7	$CH_3CH_2\ CH_2\ CH_2CH_2CH_2CH_3$
Octane	8	$CH_3CH_2\ CH_2\ CH_2CH_2CH_2CH_2CH_3$
Nonane	9	$CH_3CH_2\ CH_2\ CH_2CH_2CH_2CH_2CH_2CH_3$
Decane	10	$CH_3CH_2\ CH_2\ CH_2CH_2CH_2CH_2CH_2CH_2CH_3$

Side chains/ branches

- Carbon based substituents are named before the chain name.
- They have the prefix -**yl** added to the basic stem (e.g. CH_3 is methyl).

Name	Number of carbon atoms	Compound	Prefix
Methane	1	CH_3-	Methyl
Ethane	2	CH_3CH_2-	Ethyl
Propane	3	$CH_3CH_2CH_2$-	Propyl
Butane	4	CH_3CH_2 CH_2CH_2-	butyl

- Number the principal chain from one end so that the side chain is attached to a carbon atom with **the lowest possible number**.



e.g. 2-methylhexane

If there is more than one side-chain the following rules apply:-

- Side-chain names appear in **alphabetical order** i.e. butyl, ethyl, methyl, propyl.
- Number the principal chain from one end to give the **lowest numbers**.
- Each side-chain is given its own number.
- If **identical side-chains** appear more than once, use prefix with **di, tri, tetra** etc
- Numbers are separated from names by a HYPHEN ***2-methylheptane***
- Numbers are separated from numbers by a COMMA ***2,3-dimethylbutane***

Exercise 1: (a) Draw the **displayed** structures for the following alkanes

2-methylbutane	2-methylpropane	3-ethylpentane
2-methylpentane	3-ethylheptane	2,2-dimethylpropane

2,3,5-trimethylhexane	2,2,4-trimethylpentane	4-ethyl-2,5-dimethylheptane

Cycloalkanes-

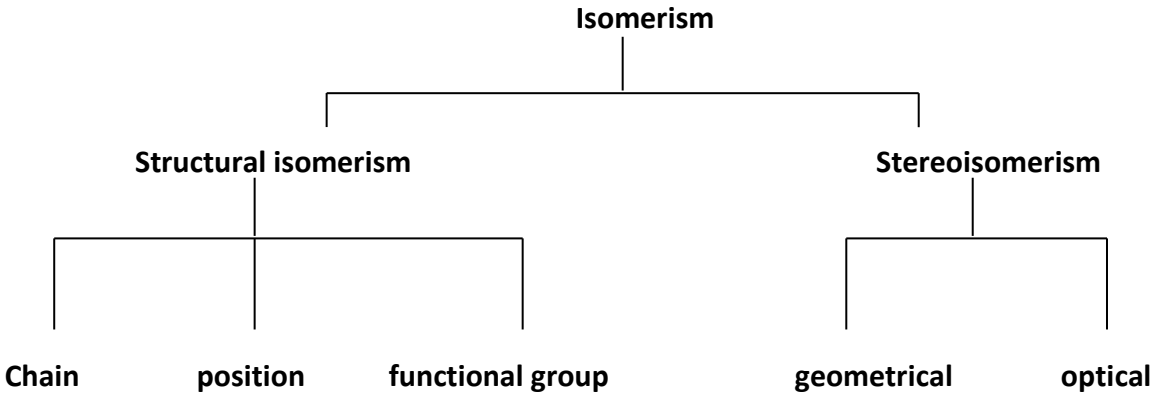
- When naming cycloalkanes, the prefix **CYCLO** is used together with the above general rules.

(b) Draw the following cycloalkanes

Cyclopropane	Cyclohexane	1,3-dimethylcyclopentane

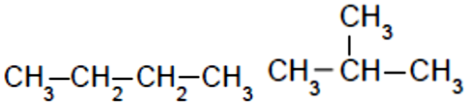
4.10 ISOMERISM

- Isomerism occurs when two or more compounds have the **same molecular formula** but have **different structural formula or different arrangement of atoms** in space.
- Isomerism can be subdivided as below

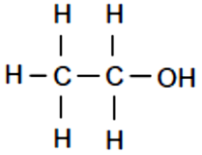


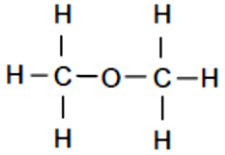
Structural isomerism: have the ***same molecular formula*** but ***different structural formulae***

- Chain isomerism: occurs due to branching in carbon chains.

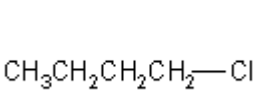


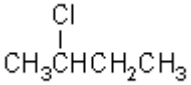
- Functional group isomerism: occurs due to the presence of different functional groups.





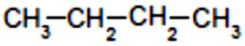
- Position isomerism: occurs due to the position in which the functional group is present.



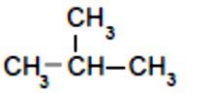


4.11 Isomerism in alkanes

- Alkanes exhibit structural isomerism which occurs due to branching in carbon chains.
- The first example of isomerism occurs with C_4H_{10}, with two isomers.



butane



2-methylpropane

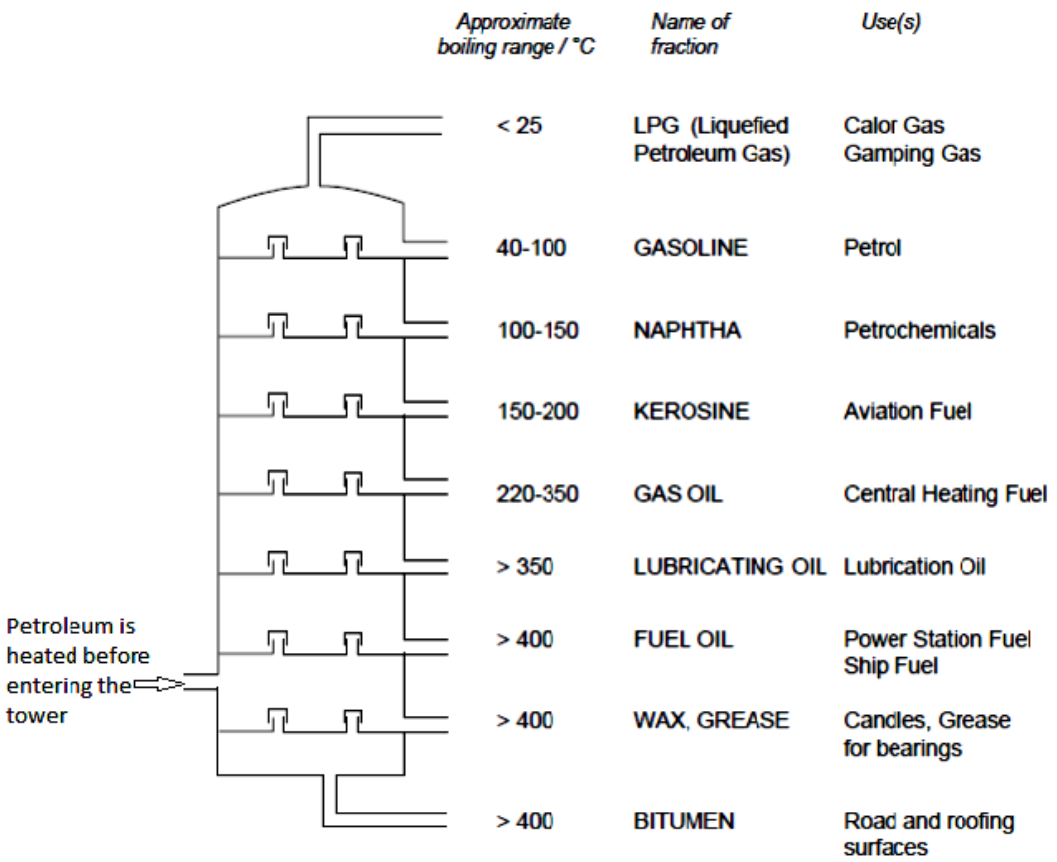
Exercise 2: Draw and name all the isomers of

(a) C_5H_{12}

(b) C_6H_{14}

4.12 The petrochemical industry (production of alkanes)

- The most important source of alkanes is fossil fuels like **crude oil** or **petroleum**.
- The composition of crude petroleum varies according to its source.
- It is a dark colored, viscous liquid that consists mostly of alkanes with up to 40 carbon atoms. It also contains some alkenes, alkynes, water, sulfur and sand.
- It can be split up into useful fractions by fractional distillation.
- Distillation separates the compounds according to their boiling point.
- At each level a mixture of compounds in a similar boiling range is taken off. It is known as a **fraction**.
- Rough fractions can then be distilled further to obtain narrower boiling ranges.
- Some fractions are more important - usually the lower boiling point ones
- High boiling fractions may be broken down into useful lower ones by **CRACKING.**
- The main fractions that result when crude oil is distilled are;
    - **Refinery Gas** :
        - Contains gaseous alkanes (C_1 to C_4) with a larger proportion of methane.
        - Used mainly as fuel or as a starting point for other organic synthesis.
    - **Gasoline:**
        - Contains a mixture of liquid hydrocarbons (C_5 to C_{10})
        - Used widely as fuels (petrol) in internal combustion engines that power cars.
        - Contains a mixture of straight and branched chain alkanes. The more branched molecules there are the more useful the fuel.
    - **Kerosene:**
        - Contains mainly C_{11} and C_{12} fractions.
        - Used largely as fuel for aircraft engines.
        - Can be cracked to form gasoline.
    - **Diesel oil/ Gas oil:**
        - Used in diesel engines and as a fuel for industrial boilers.
    - **Residue:**
        - A complex mixture of hydrocarbons that can be further fractionated to form lubricating oil, fuel oil, waxes and bitumen.

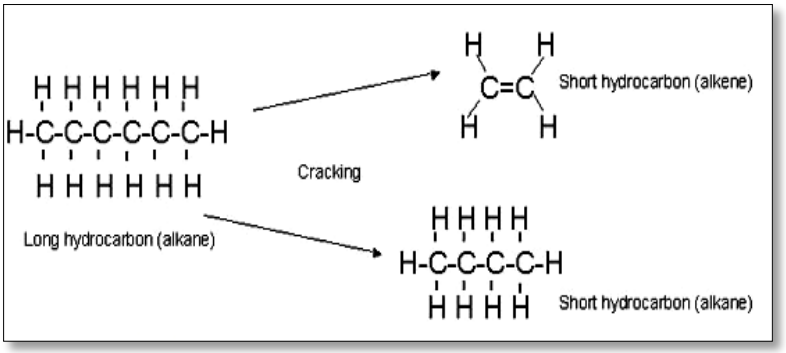


Cracking

- The demand for the lighter fractions is higher compare to the heavier ones. But the heavier fractions make up more than 50% of the products from distillation of petroleum.
- Therefore, these heavier long-chain fractions can be broken down to simpler fractions by cracking.
- There are two types of cracking; ***thermal cracking*** and ***catalytic cracking***.

Thermal cracking

- Heating the longer chain alkanes at very high temperatures (400 – 900°C) causes the molecules to split and form shorter chain molecules.
- Bonds can be broken anywhere by C-C bond fission or C-H bond fission.
- Longer chain molecules can be cracked to form either an alkene and hydrogen or an alkane and alkene

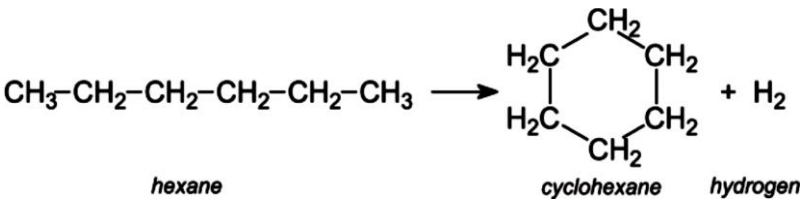


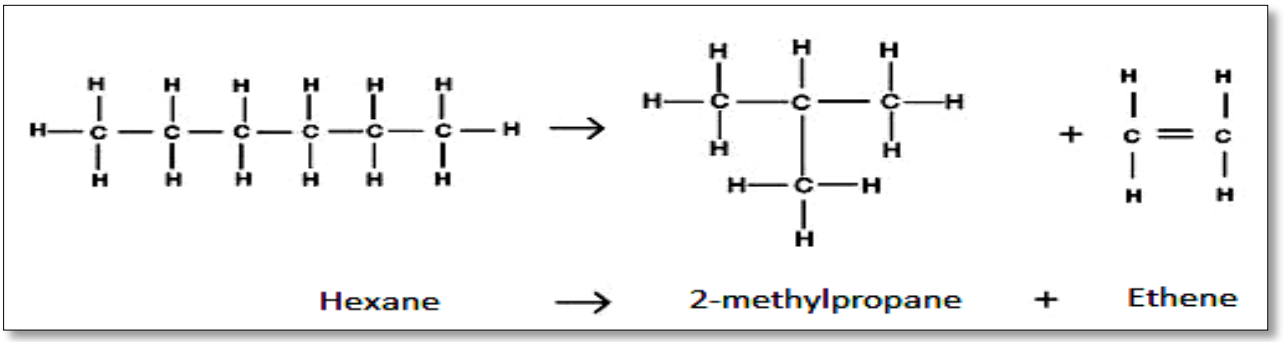
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Catalytic cracking

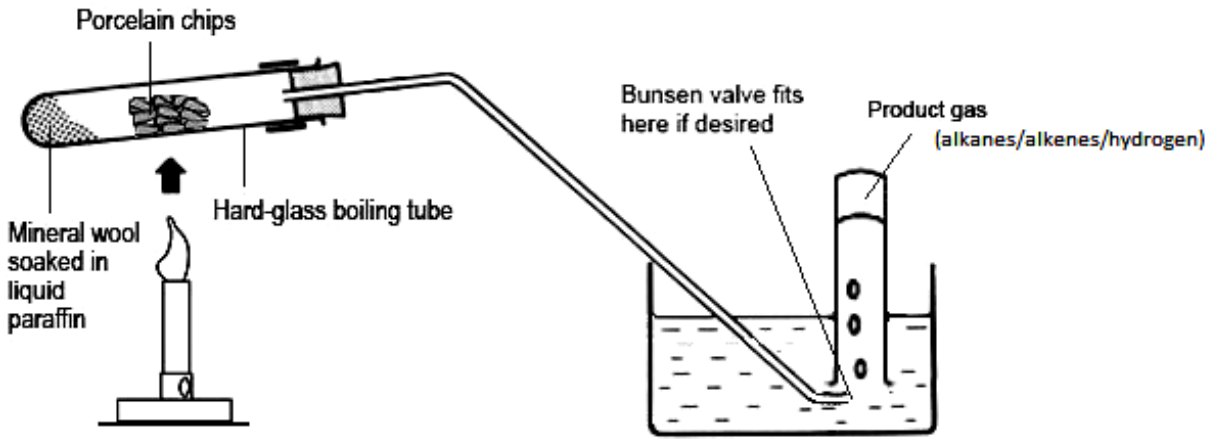
- Producing the high temperatures for cracking is very expensive.
- By using catalysts the process can be carried out at lower temperatures. It is known as ***catalytic cracking*** or ***cat cracking.***
- Zeolite (crystalline aluminosilicates) are commonly used as catalysts.
- Unlike thermal cracking, branched and cyclic alkanes can be formed by catalytic cracking. This is referred as **catalytic reforming**.

The branched alkanes are more useful as fuels as their boiling point is lower compared to the straight chain alkanes.





Appratus used in the laboratory for cracking



4.13 Combustion of alkane fuels

- Alkanes are useful fuels as they burn easily in the presence of oxygen to release energy in an exothermic reaction.
- Alkanes only burn when they are in gaseous state. So, liquid and solid alkanes are less flammable because they must be vaporized before burning.
- Methane (natural gas), propane and butane are commonly used fuels.
- Propane and butane can be liquefied under high pressure and at low temperatures to form **liquefied petroleum gas (LPG**). This is an advantage over methane because LPG can be transported easily and a large quantity of gas can be stored in a small space.

***Reagent*:** Oxygen gas

Reaction condition: A spark or ignite the mixture.

Complete combustion: $\mathbf{CH_{4(g)} + 2O_{2(g)} \rightarrow CO_{2(g)} + 2H_2O_{(l)}}$

Incomplete combustion: $\mathbf{CH_{4(g)} + 1½O_{2(g)} \rightarrow CO_{(g)} + 2H_2O_{(l)}}$

- When alkanes undergo complete combustion, carbon dioxide gas is formed.
- Carbon dioxide is a greenhouse gas which leads to global warming and climate change.
- When alkanes undergo incomplete combustion carbon monoxide is formed which is a poisonous gas.

Oxides of sulfur

- Some of the molecules in crude oil contains atoms of sulfur these may not be removed by fractional distillation.
- During the combustion of alkanes atoms of sulfur forms sulfur dioxide gas, and then react in atmosphere to form sulfur trioxide gas, when they dissolve in water in the atmosphere, they form sulfurous acid and sulfuric acid. Contribute to the formation of acid rain.

  $S+O_2 \rightarrow SO_2$ and $2SO_2+O_2 \longrightarrow 2SO_3$

  $SO_2+H_2O \rightarrow H_2SO_3$ and $SO_3 + H_2O \rightarrow H_2SO_4$

Oxides of Nitrogen

- Very few molecules used as alkane fuels contain atoms of nitrogen, their combustion occurs at very high temperatures cause nitrogen molecules in air to react with oxygen molecules. These reaction leads to the formation of oxides of nitrogen, but the main ones are (NO)

  $N_2+O_2 \longrightarrow 2NO$

  It can react with more oxygen in the atmosphere

  $2NO+O_2 \longrightarrow 2NO_2$

  Nitrogen oxide is acidic and can dissolve in water in the atmosphere, forming nitrous acid and nitric acid:

  $2NO_2+H_2O \longrightarrow HNO_2+HNO_3$

  Both acids contribute to environmental damage.

4.14 Catalytic converters to the rescue

- Cars and other road vehicles are responsible for a lot of air pollution. The wide spread use of catalytic converters fitted to the exhaust systems has made pollution less of a problem.
- There are different types of catalytic converter, but they all use small quantities of precious metals such as platinum, rhodium, palladium .These metals are spread thinly over a honeycomb mesh to increase the surface area for reaction .One common type type is known as a three-way catalyst because it can remove three different pollutants; carbonmonoxide, unburned hydrocarbon and oxides of nitrogen.

4.15 Need for alternative fuels

- Recently there have been attempts to produce new fuels as alternatives to fossil fuels most of these fuels can be described as **Biofuels**, which means that they are obtained from living matter that has died recently, rather than died many millions of years ago.
- A wide definition of biofuels would include wood, which has been used as a fuel for many centuries and also it is a renewable source.

4.16 Carbon neutrality

- Fuels can be considered in terms of their carbon neutrality. Carbon neutral is a term used to represent the idea of carbon dioxide neutrality. For example, when a tree grows, it absorbs carbon dioxide from the atmosphere, and the carbon atoms become part of the structure of the tree.
- If the tree is cut down and the wood is burned, then carbon dioxide is formed during its combustion.
- If the amount of carbon dioxide formed in the combustion is the same as the amount absorbed during the tree's growth, then the wood is described as carbon neutral.
- This is because, over the time period between the tree starting to grow and the use of its wood as a fuel, the amount of carbon dioxide in the atmosphere has not been altered by its combustion.
- Fossil fuels formed from trees, could be described as carbon neutral because they too absorbed carbon dioxide during their growth and form the same amount of carbon dioxide when they burned.

Bioalcohols

- Bioalcohols are carbon neutral fuels because they are made from recently –grown plants that have absorbed carbon dioxide from the atmosphere**.**
- The use of bioalcohols involves forming more carbon dioxide than is absorbed. Even so, bioalcohols are closer to being carbon neutral than fossil fuels**.**

Bioethanol

- Currently, the commonest bioalcohol is bioethanol. Bioethanol is identical to ethanol. The bio part of the name only refers to the method of production.
- For centuries, ethanol has been produced by the fermentation of sugars. The ethanol has to be separated from the much larger amount of water before it can be used as a fuel, and this separation requires energy.
- It is now possible to use a wide range of plants, and also plant waste to produce ethanol
- The upper limit that can be obtained from a given amount of ethanol that can be obtained from a given amount of starting material is increasing, and is much higher than in traditional fermentation. In united states, corn is the main source of ethanol used in cars.

Comparing fuels

Biofuels

- Land use-how much land is used to grow the crop? Should the land be used for other purposes, especially to grow food to feed people?
- Yield-how much of a crop can be grown on a given piece of land, and how quickly does it grow? What percentage of the carbon and hydrogen atoms in the crops ends up in the fuel?
- Manufacture and transport-how much energy is used in growing and processing and transporting the crop?
- Carbon neutrality -how close is the fuel to being carbon neutral?

Hydrogen

For a long time, hydrogen has been considered an ideal alternative fuel, it is not a biofuel. There are two main ways in which hydrogen can be used as fuel in cars:

It can be burned instead of fossil fuels such as petrol or natural gas

It can be used in a fuel cell to generate electricity that powers an electric motor.

Using a hydrogen fuel cell or burning hydrogen instead of a hydrocarbon in a car, No carbon dioxide is produced ,which suggest no increase in the greenhouse effect.

4.17 Reactions of alkanes with oxygen

- Alkanes are useful fuels as they burn easily in the presence of oxygen to release energy in an exothermic reaction.
- Alkanes only burn when they are in gaseous state. So, liquid and solid alkanes are less flammable because they must be vaporized before burning.
- Methane (natural gas), propane and butane are commonly used fuels

Exercise 3: (a) Write equations for complete combustion of

1. Butane: _____
2. Hexane: _____
3. Decane: _____

(b) Write equations for incomplete combustion of

1. Propane: _____
2. Heptane: _____
3. Octane: _____

4.17 Halogenation:

- It is the introduction of a halogen atom to an alkane.
- Alkanes react with chlorine with an input of energy in the form of sunlight or ultraviolet light.
- This is an example of a substitution reaction as hydrogen atoms are substituted by a chlorine atom.
- The chlorination of alkanes proceeds via a mechanism known as *homolytic free radical substitution*.

Chlorination of methane

Reagents - chlorine and methane

Reaction conditions - UV light or sunlight - *heat could be used as an alternative energy source*

Equations: $CH_4(g) + Cl_2(g) \rightarrow HCl(g) + CH_3Cl(g)$ *chloromethane*

$CH_3Cl(g) + Cl_2(g) \rightarrow HCl(g) + CH_2Cl_2(l)$ *dichloromethane*

$CH_2Cl_2(l) + Cl_2(g) \rightarrow HCl(g) + CHCl_3(l)$ *trichloromethane*

$CHCl_3(l) + Cl_2(g) \rightarrow HCl(g) + CCl_4(l)$ *tetrachloromethane*

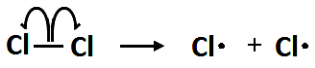
4.18 Homolytic free radical substitution

- Alkanes undergo hemolytic free radical substitution when they react with halogen.

  Example: reaction between methane and chlorine.

- This mechanism includes three steps.

  1. ***Initiation***: In the presence of sunlight (UV rays), chlorine molecules undergo homolytic fission to form chlorine free radicals.



$Cl_2 \longrightarrow 2Cl\bullet$ *radicals created*

  2. ***Propagation***: chlorine free radicals attack methane molecules resulting in a hydrogen atom being substituted by chlorine atom. This process continues in a **chain reaction**.

$Cl\bullet + CH_4 \longrightarrow CH_3\bullet + HCl$ *radicals used and*

$Cl_2 + CH_3\bullet \longrightarrow CH_3Cl + Cl\bullet$ *then re-generated*

3 Termination: Two free radicals in the system combines together to form a neutral product.

$Cl\bullet + Cl\bullet \longrightarrow Cl_2$ *radicals removed*

$Cl\bullet + CH_3\bullet \longrightarrow CH_3Cl$

$CH_3\bullet + CH_3\bullet \longrightarrow C_2H_6$

- This mechanism shows the possibility of forming hydrocarbons with more number of carbons than used as a reactant.
- Halogenation of alkenes is not an efficient method to make a specific haloalkane because the reaction cannot be controlled. There are many substitution reactions (side reactions) occurring at the same time. Hence a mixture of several products is obtained.

Exercise 4:

Write a complete mechanism when ethane reacts with I-Cl to form iodoethane.